



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the Performance of Data Analysis Students

Submitted by: Jana Islam / 24-101402

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INTRODUCTION:

Students often find it hard to stay focused while studying because of different distractions. One important factor that may affect their productivity is the study environment.

This report examines how different study environments affect students' productivity by analyzing survey responses from university students.

RESEARCH QUESTION:

How does the study environment affect students' productivity?

SAMPLING METHOD:

The sampling method used in this study is convenience sampling. The survey was distributed online through easily accessible platforms to university students. This method was selected because it allows for efficient data collection from a readily available population, making it well-suited for exploratory research on student study habits.

To improve the diversity of the sample, the survey was shared among students from different academic levels and backgrounds.

HYPOTHESIS:

Students who study in structured and quiet environments, such as libraries or desks, tend to have higher productivity levels compared to those studying in more comfortable or distracting environments like beds or noisy spaces.

POPULATION OF INTEREST:

The population of interest for this study is **university students**. This includes students from different academic years and fields of study, as they all engage in regular studying and are affected by their study environments.

BIAS IDENTIFICATION:

In designing this survey, we have taken steps to identify and minimize potential sources of bias:

- **Sampling Bias:** Most respondents (62 out of 80) were aged 18–22 and studied from home, which may not represent older students or different study environments. To minimize this, the survey was distributed as broadly as possible.
- **Response Bias:** Self-rated questions (such as comfort level and productivity) may reflect the student's mood at the time rather than their actual habits. To reduce this, all questions were kept simple, clear, and non-leading.

To reduce bias, the survey was anonymous, and neutral wording was used in all questions. Additionally, balanced answer choices were provided to avoid leading participants toward specific responses.

SURVEY QUESTIONS:

- What is your age group?
- What is your gender?
- Where do you usually study?
- On scale from 1-5 how comfortable is your study space?
- How many hours do you study per day on average?
- How often do you change your study location?
- How would you rate your productivity when you change your studying environment?
- What are the main sources of your distraction?
- In which environment are you most productive?
- How does noise in your environment affect your ability to focus?
- When are you most productive?

ONLINE SURVEY LINK:

-Link:

https://docs.google.com/forms/d/e/1FAIpQLScn_9RtwYbz6wI1SLIUDT9NEd3Zl4vsB9pRbiSgkFamISZLyQ/viewform

-Number of samples collected: 80 Responses

ANALYSIS:

-To analyse the data collected from this survey, descriptive statistical measures such as the mean, median, and mode were calculated. These measures provide a clear summary of the data and help identify overall patterns in student productivity across different study environments.

-Comfortable Study Space (rated 1–5):

- Mean: 3.75
- Median: 4.00
- Mode: 3

-Productivity Changes with Environment (rated 1–5)

- Mean: 3.41
- Median: 3.00
- Mode: 3

-Noise Affects Ability to Focus (rated 1–5):

- Mean: 3.42
- Median: 4.00
- Mode: 4

Chart 1: Productivity Levels:

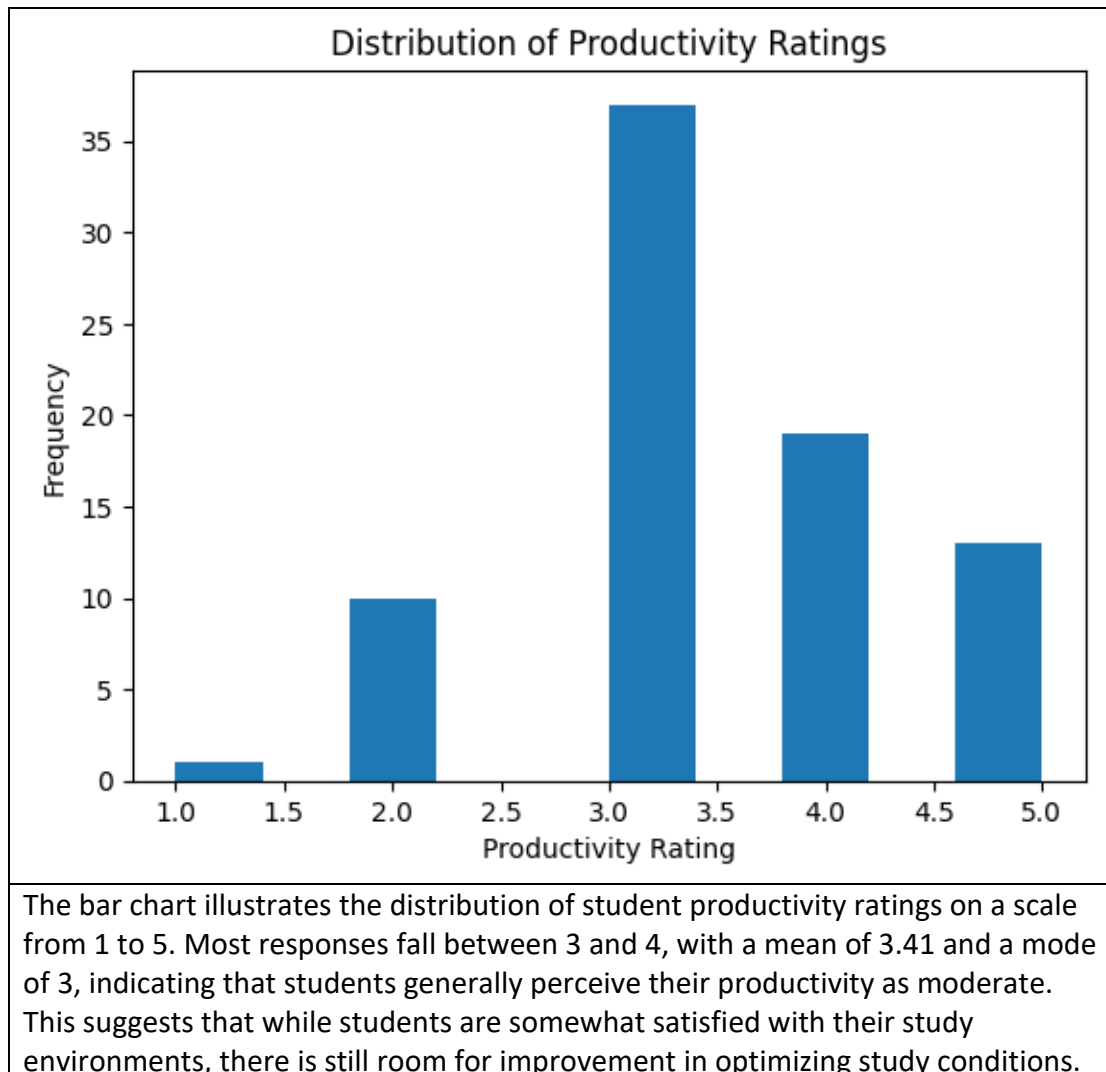


Chart 2: Sources of Distraction:

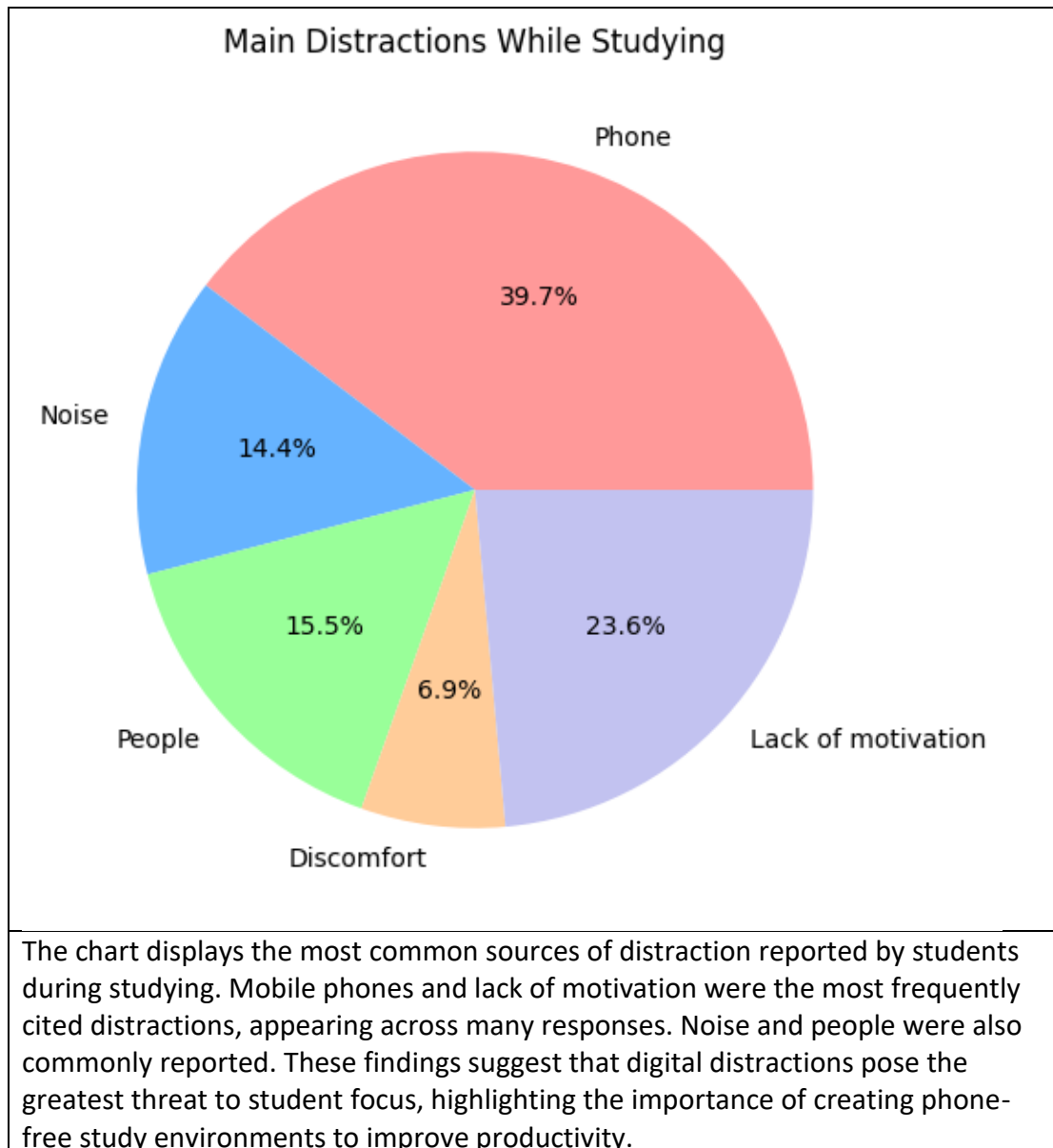
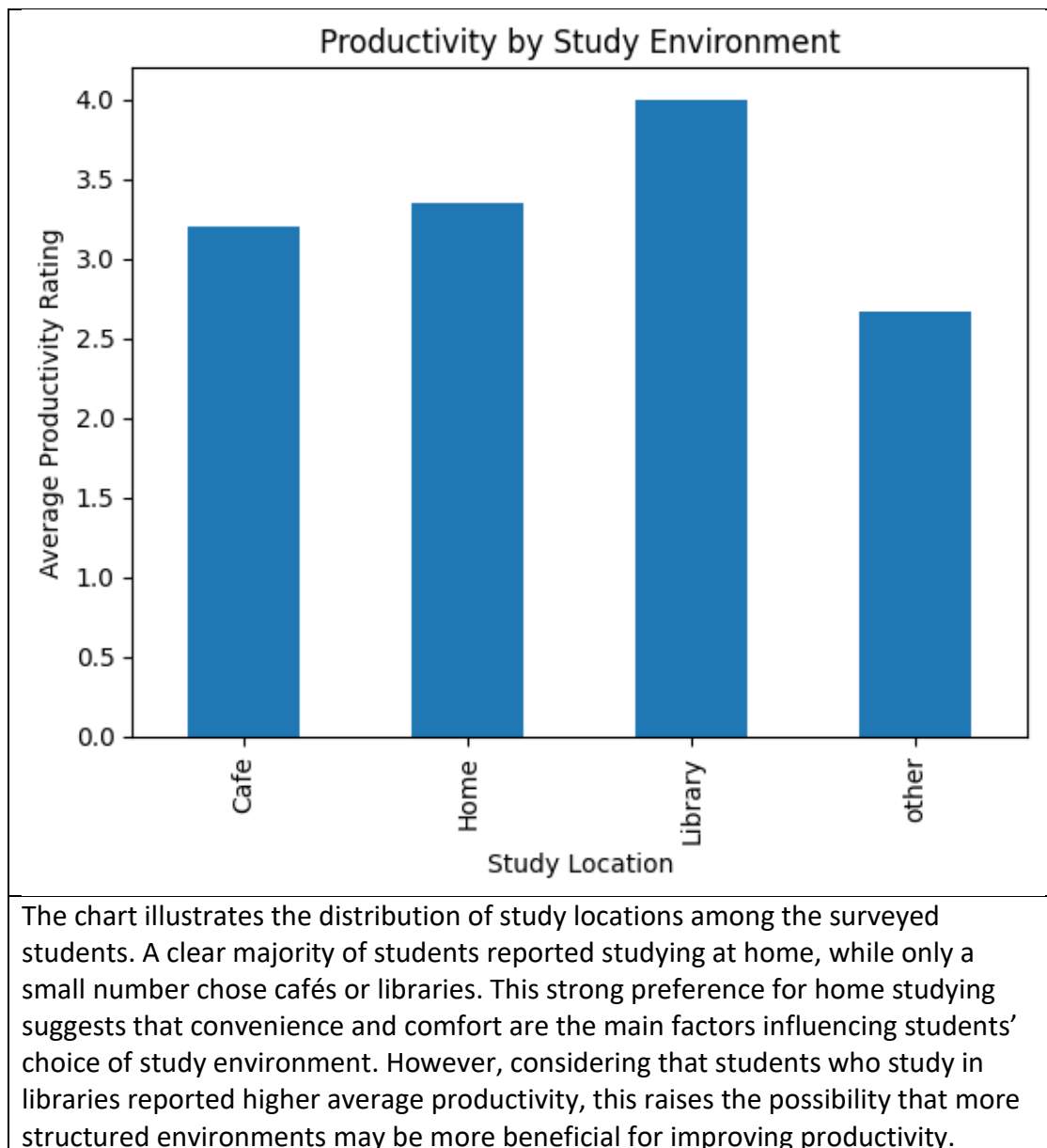


Chart 3: Productivity by Study Environment:



CONCLUSION:

Based on the analysis, there is a noticeable relationship between study environment and student productivity. Students who study in quieter and more comfortable environments tend to report higher productivity levels.

However, productivity is influenced by multiple factors, including personal habits and external distractions. Therefore, while the study environment plays an important role, it is not the only factor affecting academic performance

ANY POTENTIAL ISSUES:

Several limitations were identified during this study:

- **Limited Sample Representation:** Although 80 responses were collected, the sample is heavily skewed toward 18–22-year-old students studying from home, which may not fully represent the diversity of all university students.
- **Convenience Sampling:** The survey was distributed through easily accessible online platforms, meaning participation was voluntary and self-selected. This limits how broadly the results can be generalized to the wider student population.
- **Self-Reported Data:** All responses are based on students' own perception of their productivity and study habits rather than objective measurement. This introduces the possibility of inaccurate or exaggerated answers.
- **Question Misinterpretation:** Some participants may have interpreted certain questions differently than intended, particularly the rating-scale questions, which could affect the consistency and accuracy of the results.